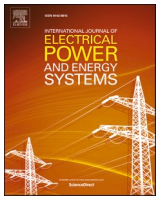
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An improved virtual synchronous generator power control strategy considering time-varying characteristics of SOC

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A B S T R A C T

Virtual synchronous generator (VSG) is proposed to solve the lack of inertia of the power system. However, as the core of the VSG providing inertia, the dynamic characteristics of energy storages tend to be ignored in the VSG. In this paper, the dynamic characteristics of energy storages are first considered into the VSG control. Based on this concept, an improved VSG power control strategy considering time-varying characteristics of state of charge (SOC) is proposed in this paper. Firstly, the dynamic characteristics are analyzed in terms of external and output characteristics. Then, Radial Basis Function (RBF) neural network is used to learn data features and handle nonlinear relationships between port voltage and output power of energy storages. The dynamic response of the VSG is improved since the dynamic indexes of the power are optimized during transients. Finally, simulation and experimental results validate the efficacy of the proposed power control strategy.

Index Terms—Energy storage, VSG, SOC, RBF neural network, Power control.

1. Introduction

In the traditional power systems, synchronous generators (SGs), with their large inertia and high damping operating characteristics, are proved to be beneficial for the stable operation of power systems [1,2]. However, in recent years, the large-scale application of new power generation technologies has gradually shaken the dominant position of SGs in the power generation side [3,4]. On the one hand, the high proportion of new energy sources connected to the grid reduces the conventional rotational inertia [5]. On the other hand, the fast response characteristics of power electronics and the random and intermittent nature of new energy sources bring large fluctuations to the grid-connected system [6,7].

One of the most promising solutions is the VSG [8], which is control algorithm for grid-connected converters that simulate the operating characteristics of SGs [9]. For SGs, the inertial potential energy comes from the rotor's rotational kinetic energy. The inertia constant is determined by the physical parameters of the SGs. A key feature of the VSG is that its parameters such as virtual damping coefficient and virtual inertia can be reasonably adjusted to improve the dynamic performance [10]. The stability of the VSG system stability is analyzed through establishing the closed-loop characteristic equation of the VSG [11]. The analysis explains the relationship between VSG parameters and system stability. Reference [8] further analyses the coupling effect between the active and reactive loops. Based on this concept, a step-by-step parameters design method is proposed to guide the design of VSG control parameters. To improve the system stability and optimize the VSG performance, a series of researches have been carried out to adjust the VSG control strategy [12–14]. For instances, bang-bang algorithm is proposed in [12] to damp the power oscillation through adjusting the inertia adaptively. According to bang-bang algorithm, the inertia is adaptively switched between a larger value and a smaller value under different conditions. Due to sudden changes in parameters, new instabilities are introduced into the system. RBF neural network with simple structure is applied in [13] to adjust virtual inertia according to the nonlinear relationship between inertia and frequency. Thanks to the favorable nonlinear mapping performance and higher parallel information processing capability of neural network, it makes the control algorithm more efficient, more intelligent and more stable compared with manual adjustment. Different from adjusting the VSG parameters directly, a fuzzy controller is augmented on the VSG topology to damp the perturbation in [14]. The system inertia is increased through adjusting the correction term added to the governor's output power, which is more in line with the physical realization of SG inertia. However, little previous research work has considered the dynamic characteristics of the source side, either in stability analysis or control design. It is clearly unrealistic that all the VSG control loops are realized by ideal DC voltage source [15].

In fact, the key to the implementation of VSG is the availability of sufficient energy buffer on the generation side for frequency regulation. The inherent uncertainty of new energy sources makes them incompatible with the energy requirements of the VSG. The authors in [16]

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studied the impact of ESS terminal resistance and voltage on its response time. It is shown that the increase in internal resistance increases the response time of the ESS. Reference [17] identifies the effect of dc-link voltage variations on ESS response dynamics and power capabilities. It is shown that the response dynamics of the reduced-capacitor ESS can be improved by adding a feedforward controller to optimize the dc-link voltage regulation. A new control system on basis of neural networks is applied to regulate the voltage and power in the ESS [18]. The proposed method can regulate the ESS power and voltage when all dynamics of ESS are unknown. As the core of VSG to provide inertia, the use of high energy density units together with high power density units increases the operational efficiency of the system [19]. In [15], a hybrid ESS consisting of a battery and an ultracapacitor is proposed to implement the function of VSG. The power of VSG is divided into inertial power and long-term power fluctuations. The ultracapacitor implements the fast-varying inertial power while the battery tackles long-term power fluctuations. However, the dynamic characteristics of the ESS is ignored in the implementation of VSG. In the hybrid ESS, voltage deviation is introduced into the control strategy with SOC feedback to extend the service life of the battery [20]. However, only the situation where the battery is charged and discharged up to the upper limit is considered. In [21], an adaptive weight is introduced into VSG control strategy to change the propensity for control according to the change of SOC. Although the process of SOC change is taken into account in VSG, SOC is still essentially used as a judgment indicator. The dynamic characteristics of the ESS are still ignored.

In this paper, the importance of the ESS in VSG control is revealed by the small signal analysis. Then, the dynamic characteristics of the battery are analyzed in terms of external and output characteristics. The analysis points out that the dynamic characteristics can be characterized by the battery current and derivative of SOC. Due to the time-varying characteristics of SOC, a complex nonlinear relationship is presented between the dynamic characteristics. Furthermore, a power control strategy considering the time-varying characteristics of SOC is proposed to improve the dynamic response of VSG. RBF neural network, with good approximation for continuous nonlinear function [22], is applied to learn the dynamic characteristics and process the nonlinear relationship.

The rest of this paper is organized as follows. In Section II, the basic principle of VSG is introduced. Then, the small signal analysis of VSG is performed. In section III, the dynamic characteristics of battery are analyzed in terms of external and output characteristics respectively. The proposed RBF neural network power control and design procedure are presented in section IV. The validity of the proposed control strategy is verified in section V. Section VI remarks the conclusion.

2. VSG control strategy and small signal model

2.1. VSG control strategy

A typical VSG control structure is shown in Fig. 1, where L_f, C_f and L_g represent inverter-side inductors, filter capacitors and net-side inductors. In this paper, the second-order model of the SGS is used as the control model of the VSG body. The equivalent rotor equation of motion is shown in equation (1):

$$\begin{cases} J\dfrac{d\omega}{dt} = T_m - T_e - D_p(\omega - \omega_0) \\ \dfrac{d\delta}{dt} = \omega - \omega_0 \end{cases} \tag{1}$$

where J and D_p represent virtual inertia and virtual damping coefficient; ω is the output angular frequency and ω_0 is the grid angular frequency; T_m and T_e represent mechanical torque and electromagnetic torque; δ is the VSG voltage angle.

The active power control loop of the VSG is shown in Fig. 2 (a). The mechanical power P_m is provided by the distributed generation systems.

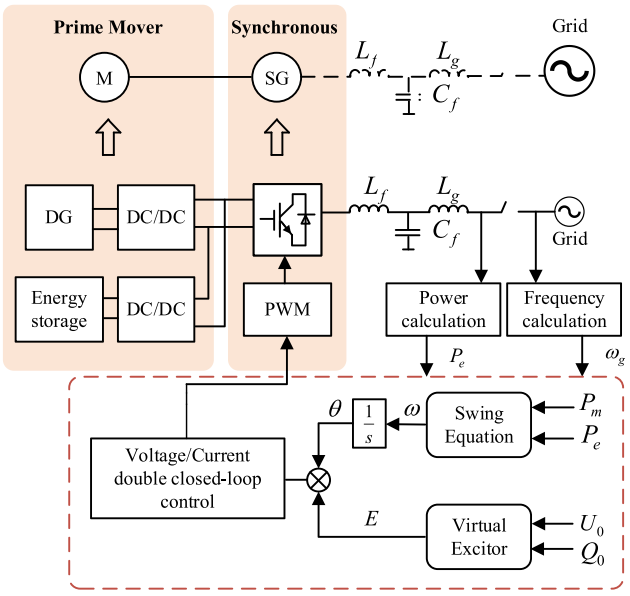


Fig. 1. Verall control block diagram of VSG.

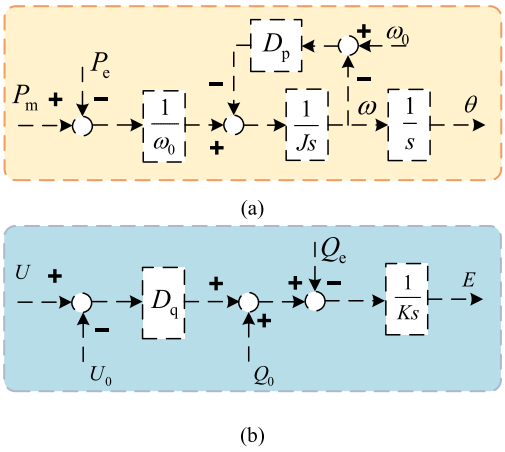


Fig. 2. Control loops of a VSG, (a) Active power control loop, (b) Reactive.

The electromagnetic power P_e is the actual output active power of the VSG.

As shown in Fig. 2 (b), reactive power loop primarily simulates the excitation equation of SGS. The excitation equation can be depicted as following form:

$$K\frac{d(E)}{dt} = Q_0 - Q_e - D_q(U_g - U_0) \tag{2}$$

where Q_0 and Q_e represent set reactive power and output reactive power respectively; U_g and U_0 represent grid voltage and reference voltage respectively; D_q and E represent the reactive power-voltage droop coefficient and the reference voltage of reactive power loop output.

2.2. Small signal analysis of VSG

The grid-connected inverter can be equated to an ideal voltage source in series with its impedance. The impedance can be depicted as following form:

$$Z = Z_o + Z_g = r + jX \tag{3}$$

where Z_o and Z_g represent inverter output impedance and grid

impedance respectively; r and X represent the resistive and inductive component of the inverter output impedance and grid impedance, respectively;

The impedance between the inverter output voltage and the grid voltage is generally resistive and inductive, and$X \gg r$. Thus, the expression for the active power fed into the grid by the inverter is:

$$P_e = \frac{EU_g \sin\delta}{X} \tag{4}$$

Since $\widehat{\delta}$ is very small, $\sin\widehat{\delta}$ is approximate to$\widehat{\delta}$, i.e.,$\sin\widehat{\delta} \approx \widehat{\delta}$. Assuming that ω_0 is nearly a constant, then$\widehat{\omega}_0 = 0$. Therefore, the active power control loop is linearized by applying the small-signal modeling, as shown in Fig. 3.

According to Fig. 3, the transfer function from $\widehat{P}_m$ to $\widehat{P}_e$ can be described as:

$$G(s) = \frac{\widehat{P}_e}{\widehat{P}_m} = \frac{EU_g}{JX\omega_0 s^2 + D_p X\omega_0 s + EU_g} \tag{5}$$

From (5), when the load reference power fluctuates in step, the expression of system output electromagnetic power is as follows:

$$\Delta P_e = G(s)\Delta P_m = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}\frac{\Delta P_m}{s} \tag{6}$$

where the expression of the natural angular frequency ω_n and damping ratio ξ are as follows:

$$\begin{cases} \omega_n = \sqrt{\dfrac{EU_g}{JX\omega_0}} \\ \xi = \dfrac{D_p}{2}\sqrt{\dfrac{X\omega_0}{JEU_g}} \end{cases} \tag{7}$$

When the damping ratio is fixed at 0.85 and the virtual inertia is varied from 0.05 to 0.5, the magnitude response curve of bode plot corresponding to (6) is shown in Fig. 4.

The output active and reactive power of ESS can be decoupled and controlled by means of a converter. The mathematical model of the ESS power conversion system can therefore be modelled by a first order inertial system as follows:

$$P_{act} = \frac{1}{1 + T_{bat}s}P_{set} \tag{8}$$

where P_{act} and P_{set} represent the actual power output of the power conversion system and set power respectively; T_{bat} can be considered as the time delay effect of ESS [23].

Considering the ESS, the small signal model of the active power control loop is shown in Fig. 5.

According to Fig. 5, the transfer function can be expressed as:

$$\begin{aligned} G_1(s) &= \frac{1}{1 + T_{bat}s} \times G(s) \\ &= \frac{\omega_n^2}{T_{bat}s^3 + (2\xi\omega_n T_{bat} + 1)s^2 + (2\xi\omega_n + \omega_n^2 T_{bat})s + \omega_n^2} \end{aligned} \tag{9}$$

Under the same conditions, the bode plot of the VSG considering ESS is shown in Fig. 6.

From Fig. 6, it can be seen that the spike in the frequency response curve proves the existence of power oscillations in the VSG dynamic process. By comparing with Fig. 4, the introduction of ESS has a significant impact on the VSG dynamic characteristics. In addition, the frequency response of the VSG with ESS is also more in line with the actual situation. Thus, it is necessary to take the dynamic characteristics of energy storage into account in the VSG control.

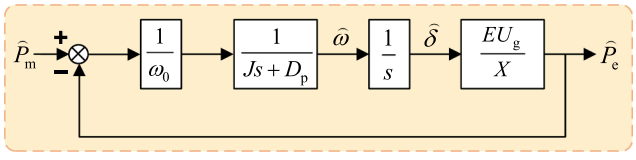


Fig. 3. Small signal model of active power loop.

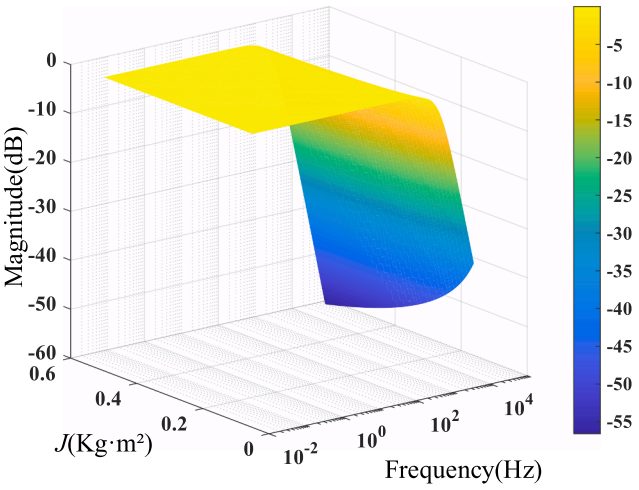


Fig. 4. Bode plot of active power loop.

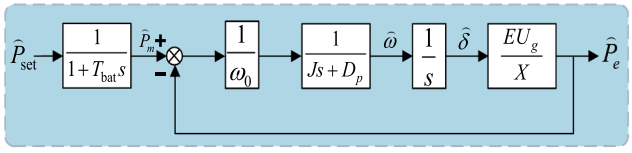


Fig. 5. Small signal model of active power loop considering ESS.

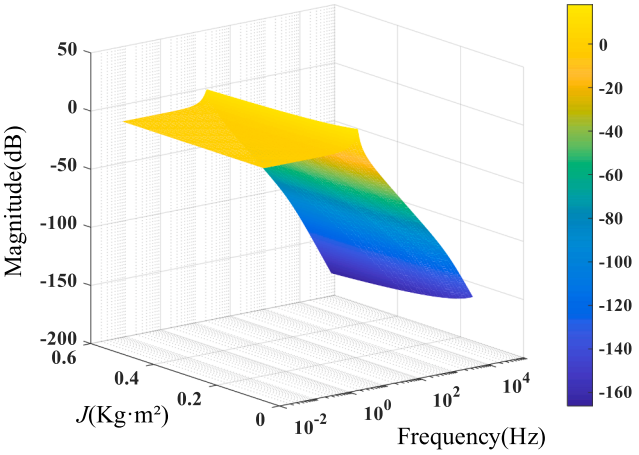


Fig. 6. Bode plot of active power loop considering ESS.

3. Dynamic characteristics of energy storages

3.1. External characteristics

Taking lead-acid batteries as an example, the variation curve of the battery port voltage with SOC is shown in the Fig. 7.

The relationship between the port voltage and SOC is obtained by means of function curve fitting as follows:

$$U_{bat}(t) = U_0 - \frac{K}{soc(t)} + A\exp[B(soc(t) - 1)] \tag{10}$$

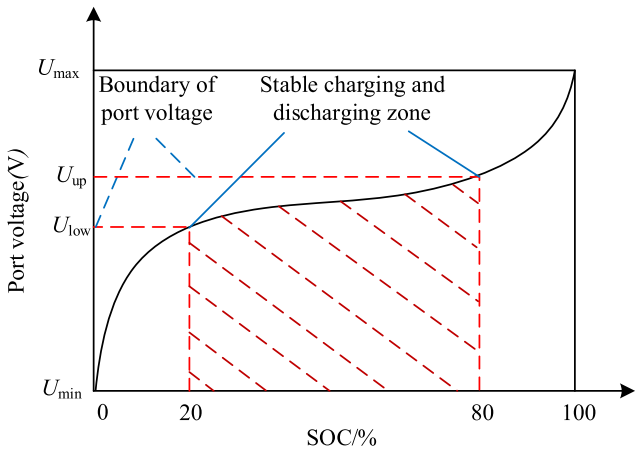


Fig. 7. External characteristics of the battery.

where U_0 is the internal electromotive force of battery; A, B and K are the corresponding curve-fitting parameters. $soc(t)$ refers to the ratio of the remaining capacity to the rated capacity under a certain charge and discharge rate,

$$soc(t) = \frac{Q_n - Q(t)}{Q_n} \tag{11}$$

where Q_n and $Q(t)$ represent the rated capacity of the battery and the amount of power released by the battery in a certain period of time, respectively.

Taking the most widely used ampere-hour integration method as an example, the calculation formula for the state of charge is:

$$soc(t) = soc_0 - \frac{1}{C_N}\int \eta I_{bat}(t)\mathrm{d}t \tag{12}$$

where soc_0 is the initial value of SOC; C_N is the rated capacity of the battery; η is coulomb efficiency coefficient; $I_{bat}(t)$ is the battery charging and discharging current.

It can be seen from (12) that SOC changes uniformly only during constant current charging and discharging. In fact, when the battery is used as a VSG DC source in response to power fluctuations, its output current is a variable determined by the external load and the internal parameters of the battery, as follows:

$$I_{bat}(t) = \frac{\Delta P}{U_{bat}(t)} = \frac{\Delta P}{U_0 - \dfrac{K}{soc(t)} + A\exp[B(soc(t)-1)]} \tag{13}$$

(13) not only shows that SOC is time-varying, but also demonstrates the coupling relationship between SOC and the battery current. According to (10) and (13), the external voltage characteristics can be characterized by the time-varying characteristics of SOC.

3.2. Output characteristics

When the power fluctuations occur in the system, VSG exhibits a virtual inertia through battery output inertial power. Setting the system for power fluctuations at 0 s and end at$t$, the cumulative energy change of the battery at time 0 ~ $t$is:

$$\Delta E_{bat} = Q_n(soc_0 - soc(t)) \tag{14}$$

The battery power output at time t is the differential of the energyΔE_{bat}.

$$P_{bat}(t) = \frac{\mathrm{d}\Delta E_{bat}}{\mathrm{d}t} = -Q_n\frac{\mathrm{d}soc(t)}{\mathrm{d}t} \tag{15}$$

It can be seen from (15) that the instantaneous power output from the battery is a correlation function of SOC derivative. The output characteristics of the battery can also be represented by the trend of the SOC variation.

3.3. VSG power control analysis

As shown in Fig. 8, the mechanical power output by the prime mover drives the rotor of the SGs to output electromagnetic power. When power fluctuations occur in the system, SGs exhibit inertia by releasing the kinetic energy stored in the rotor. While VSG represents inertia through the inertial power provided by the battery,

$$\Delta P = P_{bat} \tag{16}$$

As can be seen from (10) and (15), neither the external nor the output characteristics of the battery can be equated to an ideal voltage source. Thus, ignoring the dynamic characteristics of the battery will cause frequency fluctuations and a further increase in the differenceΔP_{vsg}, which can be calculated from (1) as follows:

$$\Delta P_{vsg} = P_m - P_e = J\omega_0\frac{d\omega}{dt} + D_p\omega_0(\omega - \omega_0) \tag{17}$$

In order to optimize the performance of the VSG, it is feasible to take the dynamic characteristics of the battery into account in the VSG power control. According to (10) and (15), the external characteristics and output characteristics of the battery can be characterized by $\dot{soc}(t)$ and $I_{bat}(t)$. Therefore, in order to solve the above issues, the dynamic characteristics of the battery considered into the VSG mechanical power can be expressed in the following form:

$$P_m = P_{ref} + P_{soc} = P_{ref} + F\left(\frac{\mathrm{d}soc}{\mathrm{d}t}, I_{bat}\right) \tag{18}$$

Facing the complex systems, it should be recognized that the key is to find out the nonlinear function$F$. The concept of artificial intelligence presents powerful control techniques to cope with nonlinear systems. By employing input–output data based on the original mathematical description of the system, RBF neural network can learn data features very well. It can handle these nonlinearities and improves the performance of the system effectively. Thus, in this paper, RBF neural network is used to fit the nonlinear function relationship between $\dot{soc}(t)$ and $I_{bat}(t)$.

4. RBF neural network controller implementation

4.1. RBF neural network model

As an efficient feed-forward neural network, RBF neural networks can map arbitrarily complex nonlinear relations. And due to its local approximation learning method, its training is more concise and convergence is faster. As shown in Fig. 9, RBF neural network mainly consists of three layers: an input layer, a hidden layer for nonlinear mapping and an output layer for linear mapping.

As shown in Fig. 9, the input of the input layer is:

$$\vec{x} = [x_1, x_2\cdots x_j]^T, j = 1, 2\cdots n \tag{19}$$

The hidden layer maps the input signal to a new space via the radial basis function, and the output of the i th hidden layer node is:

$$h_i = \exp\left(-\frac{\|\vec{x} - \vec{c_i}\|}{2b_i^2}\right), i = 1, 2\cdots m \tag{20}$$

where $\vec{c_i} = [c_{i1}, \cdots, c_{in}]$ is the center vectors of the i th hidden layer node; b_i is width of gauss function of the i th hidden layer.

The output of the output layer is:

$$y = w_1h_1 + w_2h_2\cdots w_mh_m \tag{21}$$

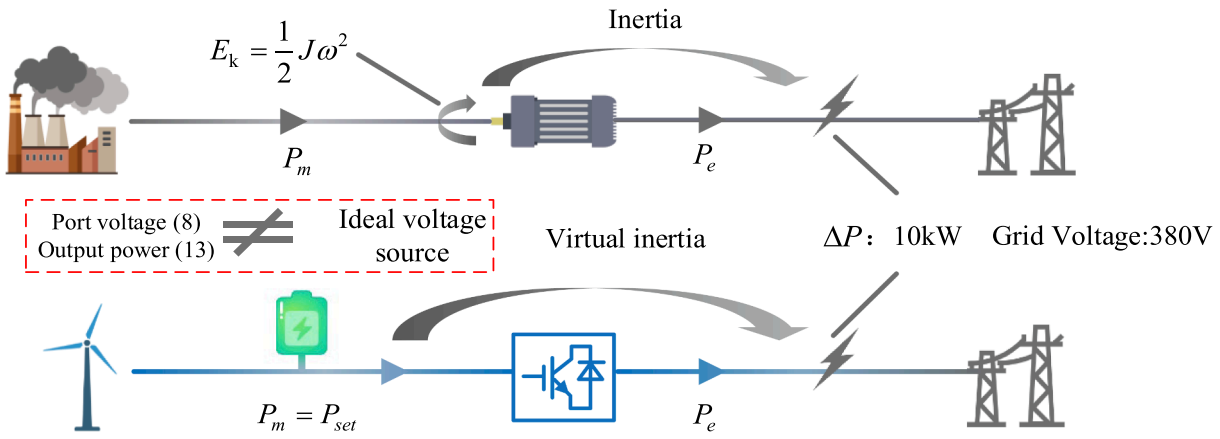


Fig. 8. Operating mechanism of VSG and SG.

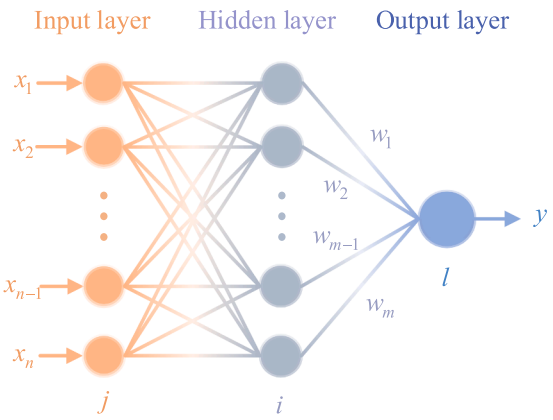


Fig. 9. RBF neural network structure diagram.

where $\vec{w} = [w_1, \cdots, w_m]$ represents the weight of the network.

4.2. Adaptive power control

Fig. 10 shows the proposed adaptive power control. Battery current and SOC are measured online to be used as the input to the neural network.

$$net_i^{(1)} = x(j), j = 1, 2 \tag{22}$$

$$x(1) = I_{bat}(t), x(2) = \mathrm{d}soc(t)/\mathrm{d}t$$

The output of the neural network input layer nodes is:

$$O_j^{(1)} = \frac{2a\tan(x(j))}{\pi}, j = 1, 2 \tag{23}$$

The input of the neural network hidden layer nodes is:

$$net_i^{(2)} = (O_1^{(1)}, O_2^{(1)}) \tag{24}$$

where superscripts (1), (2) and (3) represent input layer, hidden layer and output layer, respectively.

The output of the hidden layer nodes is:

$$O_i^{(2)} = \exp\left(-\frac{\left\|net_i^{(2)} - \vec{c_i}\right\|}{2b_i^2}\right), i = 1, 2 \cdots 5 \tag{25}$$

Linear weighting of hidden layers is:

$$net_i^{(3)}(k) = \sum_{i=1}^{5} w_{li}^{(3)} O_i^{(2)}(k) \tag{26}$$

After the activation function, the final output of the neural network is:

$$P_{soc} = O_i^{(3)}(k) = \tanh(net_l^{(3)}(k)), l = 1 \tag{27}$$

The correction term P_{soc} is added as a power adjustment signal to the mechanical power P_m in swing equation to support the inertia response of the VSG. The error term of P_{ref}-P_e is used as a performance function to adjust the weight of the neural network.

$$E(k) = \frac{1}{2}(P_{ref} - P_e(k))^2 \tag{28}$$

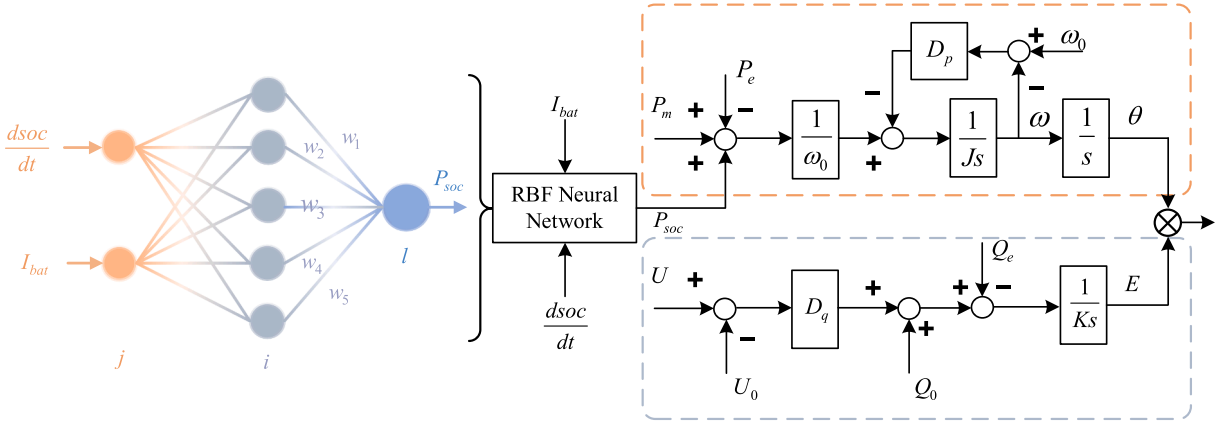


Fig. 10. The control block diagram of the proposed control.

The formula for the weight adjustment is:

$$\Delta w_{li}^{(3)}(k) = -\eta \frac{\partial E(k)}{\partial w_{li}^{(3)}} + \alpha \Delta w_{li}^{(3)}(k-1) \tag{29}$$

The gradient of performance function can be obtained as:

$$\frac{\partial E(k)}{\partial w_{li}^{(3)}} = \frac{\partial E(k)}{\partial P_e(k)} \bullet \frac{\partial P_e(k)}{\partial P_{soc}(k)} \bullet \frac{\partial P_{soc}(k)}{\partial net_i^{(3)}(k)} \bullet \frac{\partial net_i^{(3)}(k)}{\partial w_{li}^{(3)}} \tag{30}$$

In summary, the flow chart of the adaptive power control is shown in the Fig. 11.

5. Simulation verification

In order to verify the validity and effectiveness of the theories and control algorithms proposed in this paper, the model shown in Fig. 1 is established and simulated in Matlab/Simulink. The parameters of VSG and control are the same as those listed in Table 1..

5.1. Validation of battery characteristics

In this scenario, the goal is to show the dynamic characteristics of the battery in VSG system, which is different from ideal power. The simulation system is connected to the power grid. As shown in Table 1, the output active power suddenly increases by 10 kW at 0.5 s, which is supplied by the battery. After increasing the system output power, the dynamic characteristics of the battery are presented in Fig. 12.

At the beginning, the SOC of the battery drops from 50 % to 39.3 % to supply the increased power. Fig. 12(a) illustrates the dynamic response of the battery through the battery voltage compared with the ideal power. Within 0.5–1 s, the battery voltage drops from 262 V to 225.8 V, while the ideal power voltage is constant at 250 V. After entering a stable state, there is a slow increase trend in the change of the battery voltage. As shown in Fig. 12(b), the change of active power at 0.5 s results in the fluctuation of the battery current. Within 0.5–0.56 s, the output power of VSG is far away from the required active power, and the current of the battery increases rapidly to supply the required power. The overshoot of the battery and ideal power are 50.9A and 52.1A,

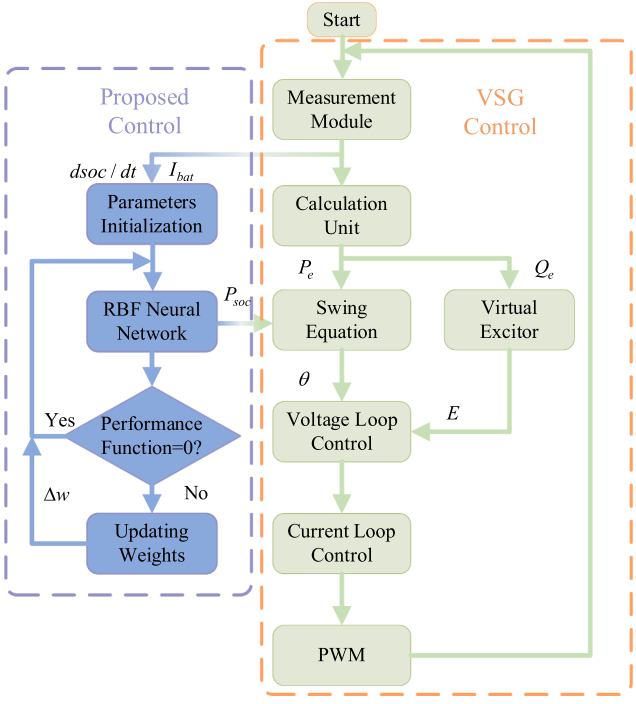


Fig. 11. Flow chart of adaptive power control.

Table 1
Electrical and Control Parameters for the Simulation.

Parameters	Symbol	Value
DC voltage	U_{dc}	800 V
Grid voltage	U_g	380 V
Ideal power voltage	U_{idl}	250 V
Grid angular frequency	ω_g	314 rad/s
Filter inductor	L_f	0.6mH
Filter capacitor	C_f	15μF
Grid side inductor	L_g	1.4mH
Virtual inertia	J	0.25 kg • m^2
Virtual damping coefficient	D_p	25 N • s/m
Learning rate of neural network	η	0.5
Inertial coefficient of neural network	α	0.05
Load increment	ΔP	10 kW
Battery capacity	Q_N	525mAh
Initial SOC	SOC	50 %

respectively. During 0.56–1 s, the output power of VSG is close to the rated value, and the current enters steady state after oscillation. The battery current is still increasing slowly after entering the steady-state due to changes in battery voltage. As can be seen in Fig, 12(c) and (d), the power and frequency overshoots of the battery are higher than the ideal power due to fluctuations in voltage and current.

5.2. Validation of adaptive power control

In this scenario, the effectiveness of the proposed RBF-VSG to a load change as a common disturbance is investigated. The results of proposed control under imbalance of power supply and demand compared with common VSG control are presented in Fig. 13.

The results of RBF neural network calculations are given in Fig. 13 (a). During 0.5 s to 0.56 s, in order to reduce the impact of changes in the dynamic characteristics of the battery on the VSG, P_{soc} first decreases to a negative value, which slows down the change of SOC and the change of VSG output power. In addition, taking into account the dynamic response of the system, P_{soc} increases rapidly to 0 in a relatively short period of time. At 0.56 s, the output active power reaches the set value for the first time, and the value of P_{soc} is 0. Then the system enters the overshoot process, and P_{soc} is negative to reduce the overshoot of the system and converge quickly. After the system has entered a steady state, the value of P_{soc} returns to 0 and does not change. As shown in Fig. 13(b), the overshoot of the output active power of the proposed RBF-VSG and conventional VSG are 1.11 kW and 2.63 kW, respectively. It is obvious that the proposed control has not only a better power dynamic response but also better inertia characteristics. Fig. 13(c) shows the frequency response curve of the system connected to the grid including RBF neural networks controller and basic VSG controller. It can be observed that the frequency maximum of the RBF-VSG and conventional VSG are 315.05 and 315.3 rad/s, and the frequency nadir are 314.02 and 313.84 rad/s respectively. It is obvious that the maximum deviation of frequency is smaller and the dynamic response is better compared with the conventional VSG.

6. Experimental results

In order to access the effectiveness and appropriate performance of the proposed RBF-VSG, experimental tests are performed on the RT-LAB OP5700 platform shown in the Fig. 14. The RBF-VSG control platform is coded from Simulink environment to the RT-LAB system. The DSP model used in the controller is TMS32F28335. The proposed method in this paper is implemented on the DSP. The voltage and current signals generated by the RT-LAB are passed to the DSP. Then DSP generates PWM signals to the RT-LAB after calculation to form a closed loop.

The parameters of the used experimental setup have been matched to those employed in simulations. Therefore, it is feasible to compare the results of simulation and experiment. Fig. 15 shows the experimental

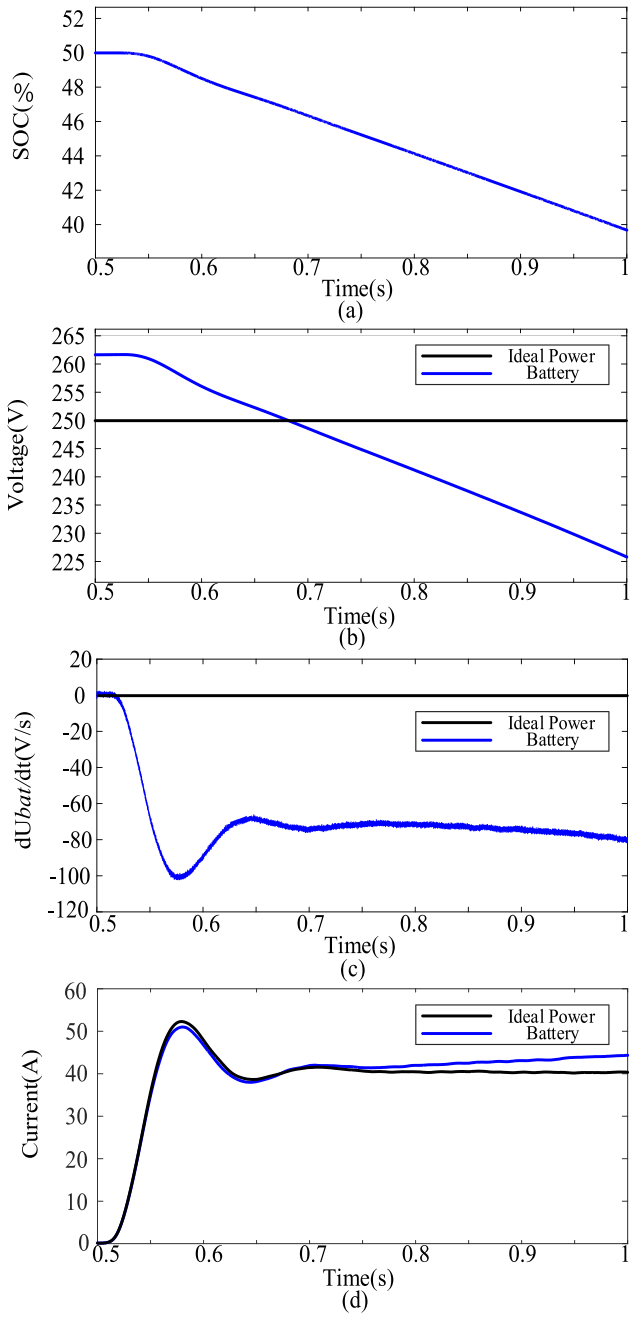


Fig. 12. Results of the battery dynamic characteristics. (a) Voltage of different situations. (b) Current of different situations. (c) Active power of different situations. (d) Angular frequency of different situations.

outcomes of VSG and RBF-VSG. Simulations and experiments match well with each other. The dynamic response of the VSG can be effectively optimized in the course of the experiment. The demonstrated agreement in simulations and experiments reveals the effectiveness of the proposed control for VSG.

7. Conclusion

In this paper, the dynamic characteristic of the battery is considered into VSG control. RBF neural network is applied to learn the data features of the dynamic characteristic and deal with the nonlinear relationships between SOC and battery current. Simulation results demonstrate that the proposed strategy has good performance in improving the dynamic performance of the system. The main work and

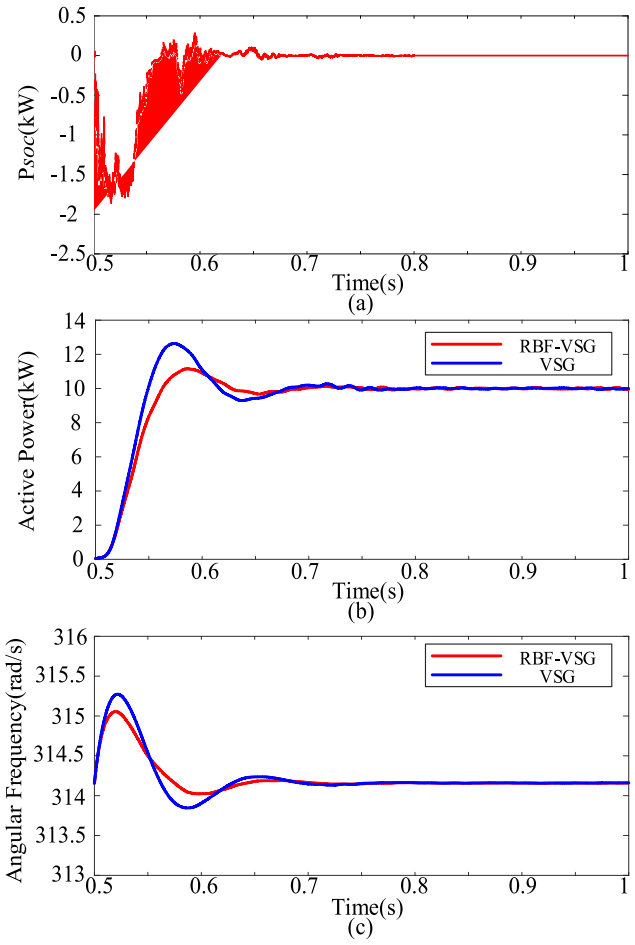


Fig. 13. Results under imbalance of power. (a) Output of the proposed control strategy. (b) Active power of different control strategies. (c) Angular frequency of different control strategies.

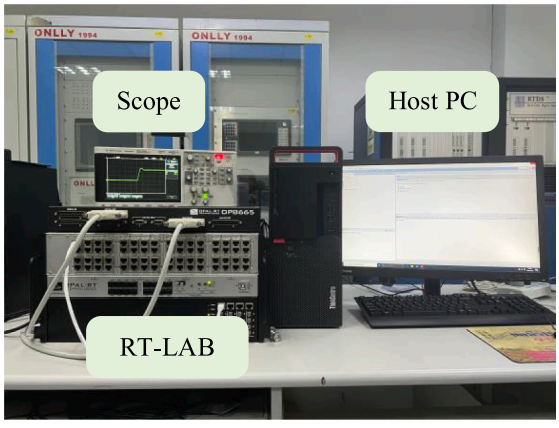


Fig. 14. Experiment Setup.

conclusions of this paper are as follows.

(1) According to the small signal analysis of VSG system, the introduction of ESS leads to power oscillations in VSG control. The response characteristics of the VSG considering ESS are more in line with reality. It is necessary to take the dynamic characteristics of the battery into account in the control of VSG.
(2) The dynamic characteristics are analyzed in terms of external and output characteristics. The analysis points out that the dynamic

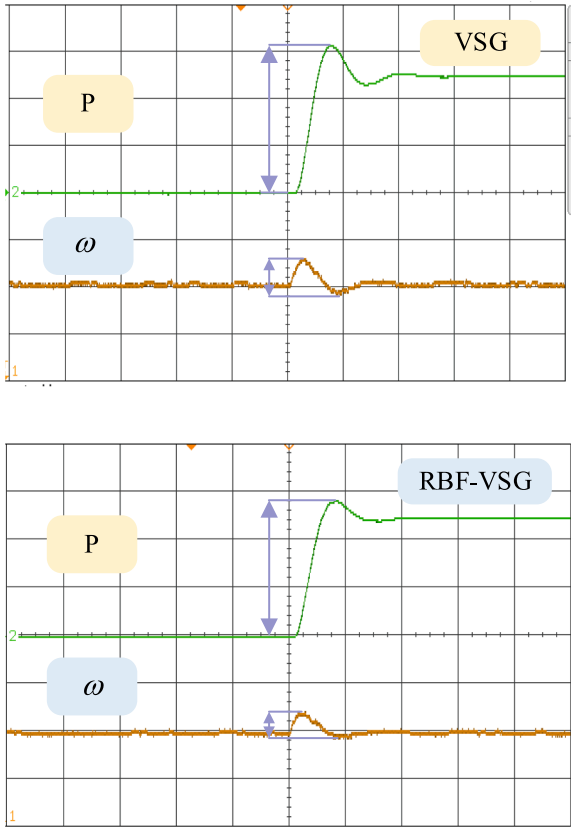


Fig. 15. Experimental results of VSG and RBF-VSG.

characteristics can be characterized by the time-varying characteristics of SOC.

(3) The mechanism of VSG power control is analyzed from the energy point of view, and an adaptive power control strategy based on RBF neural network is proposed. The proposed strategy can learn dynamic characteristics efficiently and improve VSG system performance.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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